Deforestation

A common concern we hear is that if everybody used woodgas, the trees would all be gone. While this is plausible at first glance, a closer look reveals that there are far more trees growing than we have uses for.

Scrap wood

Woodgas is usually made from waste wood. Chips, sawmill slabs, off-cuts, slash, and tree trimmings can all provide plenty of fuel without felling a tree. In fact, there is so much extra wood around that it is common practice to burn it off - simply to get it out of the way. Anyone in the country has seen burning brush piles. Any gasification from these fuel sources is not contributing to deforestation, and may well be reducing smoke pollution

Hard Work

Whether or not there's enough wood, it may be a moot point. Woodgas is such a labor intensive fuel that only a few enthusiasts are willing to use it. Chopping wood, cleaning ashes out, stopping to refuel, dealing with reduced performance - these inconveniences will keep gasification from becoming mainstream so long as cheap gasoline is available. In a petroleum shortage, more people will be receptive to woodgas, and usage will go up dramatically.

How Much Wood?

Wayne says:

"Most people want to know, how much wood does it burn? In controlled studies we did at Auburn University, my Dodge Dakota got about 1.25 miles per pound of dry wood. It turns out that the truck is 37% more efficient on woodgas than gasoline, so it works out to 16 lbs per gallon equivalent.

"Sometimes people jokingly ask how many miles I get per log. When I tell them that my truck goes 5,200 miles per cord they stop laughing. Firewood in my (rural) area sells for around $50 per cord. So if I were to buy wood, I could travel for less than a penny per mile. However I have never had to buy firewood or cut a live tree to feed my trucks. I get all the scrap wood I can use from my homemade sawmill. Wood gas is a great fit for me since I have scrap wood laying right in my way."

Run the Numbers

We propose that it's possible to provide more than enough wood for passenger vehicles in this country, even with conservative forest management. Here's some figuring we did to determine if this is feasible (references linked):

- In 2004, there were 199 million drivers[1] (http://www.infoplease.com/ipa/A0908125.html) on the road (passenger vehicles only).
- The average driver goes about 13,500 miles per year[2] (http://www.fhwa.dot.gov/ohim/onh00/bar8.htm).
- In 2004, average fuel economy was 20 mpg[3] (http://www.epa.gov/fueleconomy/fetrends/1975-2014/420s14001.pdf) (16 lbs/gal = 0.8 lbs/mile on wood)
- Therefore, you would need about 10,800 lbs per driver (5.4 dry tons).
- Total wood requirements = 199 million x 10,800 pounds = 2.1 trillion pounds or 1 billion dry tons.
- An acre of woodland can produce at least 1.5 cords/3 dry tons of wood annually (very conservative) - need 333 million acres.
- The national forest is estimated at 747 million acres[4] (http://www.fs.fed.us/nrs/pubs/gtr/gtr_wo78.pdf). The contiguous US is 1.89 billion acres; 40% forested.
- The US farmland devoted to corn for ethanol was estimated in 2011 at 35 million acres (13.9 billion gallons[5] (https://en.wikipedia.org/wiki/Ethanol_fuel_in_the_United_States)/400 gal per acre[6] (https://en.wikipedia.org/wiki/Ethanol_fuel#Sources)). Currently serves about 20.6 million drivers on ethanol, not counting energy inputs and soil depletion.
- Planted in trees it would yield 315 million tons of wood, managed for max yields at 9 dry tons/acre. This would supply enough wood for 58 million woodgas drivers.
- Current "readily available" logging waste is about 49 million dry tons per year[7] (/library/wood-resources/). This can fuel about 9 million woodgas drivers.
- Currently (2015) there are only 100 woodgas vehicles on the road in the US, many of them low mileage. Estimated total woodgas vehicle consumption is under 300 tons per year, all scrap wood.

Doug Brethower says:

"Couple of comments on running out of trees.

"I visited Lied Lodge last August to see their 10 tpd biomass energy system for heating and cooling their showcase lodging for the National Arbor Days Foundation in Nebraska City, NE.

"The original vision was coppicing an acreage of trees to feed the system. After 21 years of operation, they continue to get local waste wood for far less than the labor cost of coppicing trees right on the grounds.

"I asked Jason, the engineer that runs the system, his take on woodgas vehicles. He replied that it is the view of the Arbor Days Foundation that trees are a valuable resource, to be used to improve the condition of humanity in every way possible. "When you use a tree, plant two".

"Biomass energy saves Lied Lodge 60-80% on energy costs v similar size lodging. Their system also passed the latest emissions testing last year.

"Another personal short take, I was flying around the Western US a lot back when Yellowstone caught fire and other large forest fires were burning.

"Millions of acres of trees left unmanaged are a fire hazard. All that energy goes up in smoke with the many hazards of uncontrolled combustion when the inevitable occurs. The drought in the west is not likely to improve the situation."

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About Drive On Wood

A community of gasification enthusiasts hosted by Wayne Keith, resident expert & woodgas pioneer.

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